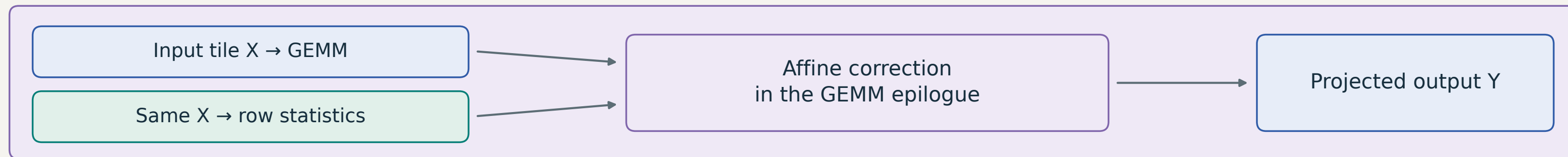


LayerNormLinear: normalize through the GEMM epilogue

CuTe SM90 main-kernel algorithm [C7] · weight preparation is separate

A / EXECUTION AND FUSION



One main kernel. No full normalized activation is materialized in HBM.

B / ALGORITHM AND DATA REUSE

ALGEBRAIC REARRANGEMENT

$$Y = r[X(\gamma W) - \mu(\mathbf{1}^T \gamma W)] + \beta W$$

GEMM and row statistics use the same input tile.
 Mean / reciprocal standard deviation correct the output.
 Weight-side terms are prepared outside the main kernel.

The diagram describes CuTe; the portable timing is a different path.

HISTORICAL MEASUREMENT

Portable Triton reference ONLY; not the CuTe path

L384

0.350 → 0.169 ms

2.07x

Baseline: Pre-change portable Triton

Shape: D128 to 16

The portable reference does not quantify the CUDA/CuTe fusion shown here. A matched CuTe comparison remains pending.